

EngiLab Frame.2D 2022 Analysis Report

Model: "Model2.fr2d"

Report created: Tuesday, 09 January 2024, 11:40:39

Node Displacements

Node ID	X-Displ. [mm]	Y-Displ. [mm]	Z-Rot. [deg]
1 (A)	0.00	0.00	0.03
2 (B)	0.00	0.00	-0.03
3 (C)	-0.09	-16.12	-0.32
4 (D)	-0.09	0.00	-0.04

Support Reactions

Node ID	FX-Force [kN]	FY-Force [kN]	MZ-Moment [kN·m]
1 (A)	12.29	3.64	
2 (B)	-12.29		

Background color: Free DOF Constrained DOF Spring DOF

Element End Forces

Element ID	Axial F. [kN]	Shear F. [kN]	Bend. M. [kN·m]
1 (A) Start	0.00	3.64	-12.29
1 (A) End	0.00	2.50	0.00
2 (B) Start	-3.64	-12.29	0.00
2 (B) End	-3.64	-12.29	-10.44
3 (C) Start	0.00	-12.29	1.84
3 (C) End	0.00	-12.29	0.00

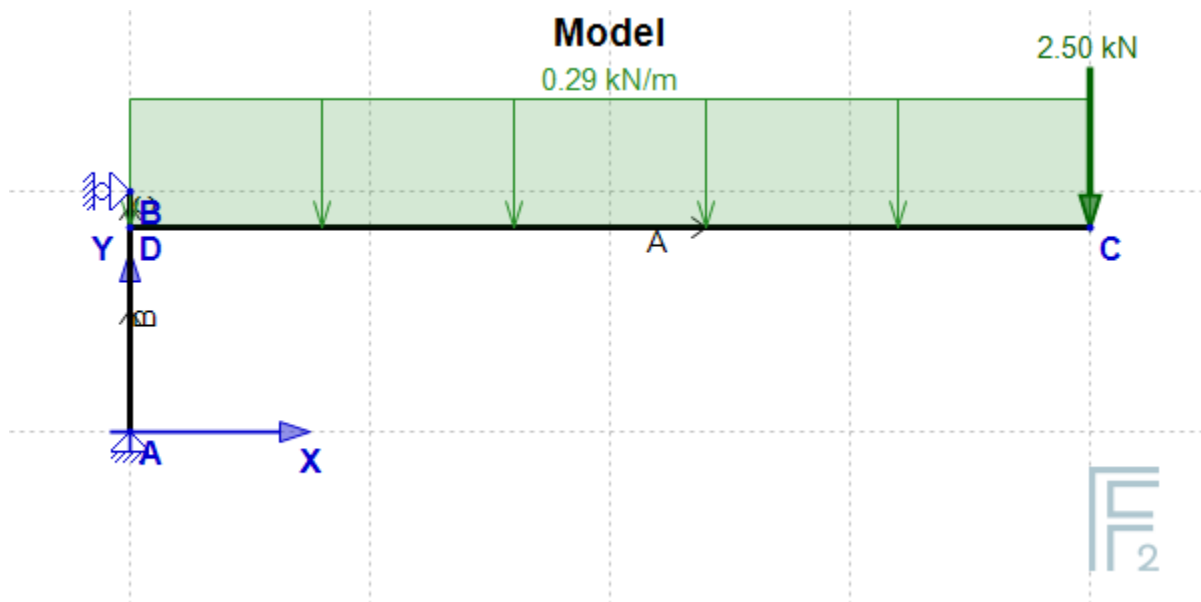
Background color: Normal DOF Hinge DOF

Sign convention: According to the diagrams' sign convention

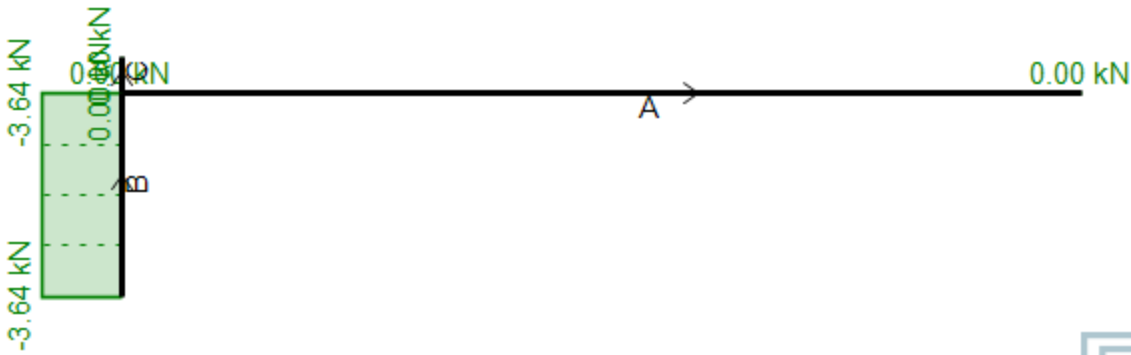
Axial Force: Positive when member is in tension

Shear Force: Positive when it rotates the member clockwise

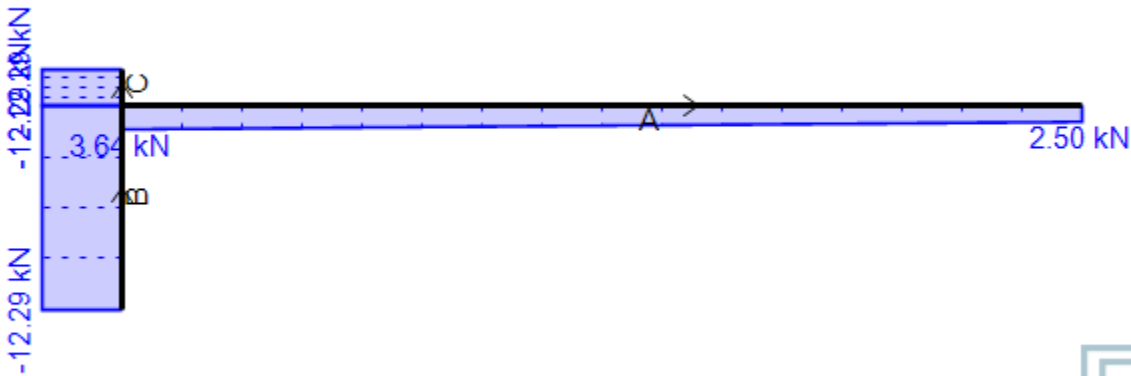
Bending Moment: Positive when the "bottom" fiber is in tension ("top" fiber in compression)



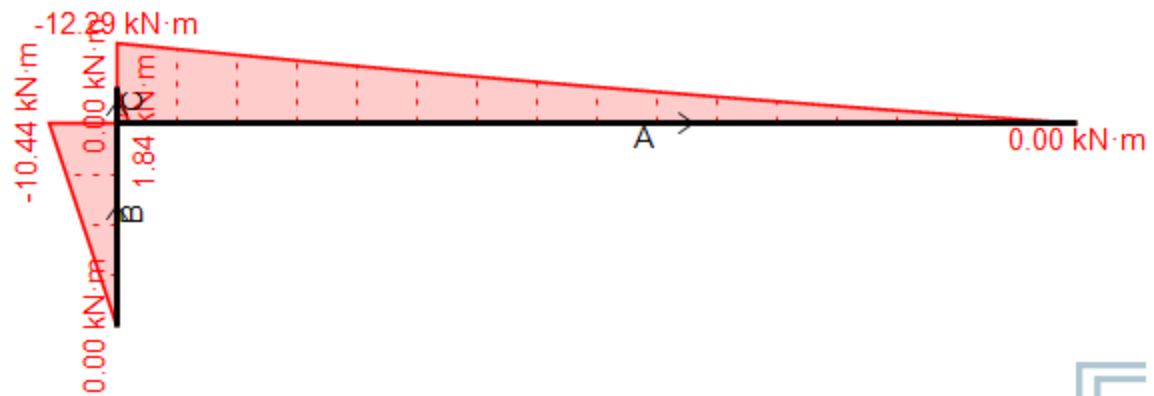
Axial Force Diagram [N]



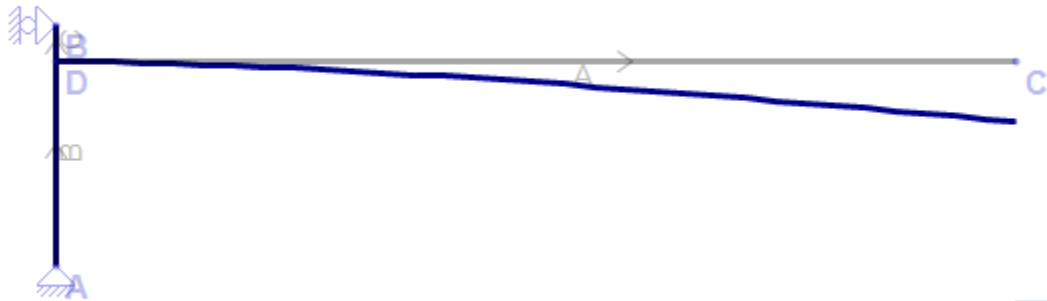
Shear Force Diagram [V]



Bending Moment Diagram [M]



Deformation (x15.51)



Free Body Diagram

